



CROP2GO

AI-Integrated Digital FPO Platform: Farm-to-Fork Traceability, Computer Vision Quality Grading, and Shared Demand Logistics

Live Prototype: <http://localhost:5173> | Backend API: <http://localhost:3001/api> | Database: SQLite Engine with Persistent Telemetry Stream

4 Roles

FARMER • FPO • BUYER •
TRANSPORTER

< 3 Sec

AI QUALITY GRADING
(A/B/C/REJ)

42%

Savings

SHARED VILLAGE
TRANSPORT LOGISTICS

100% Audit

TRANSPARENT FARMER
ESCROW PAYOUTS



Section 1: The Winning Pitch Structure for the SIH Jury

The 30-Second Elevator Hook (Say this immediately when your demo starts)

"Respected Jury Members, in India today, 140 million smallholder farmers lose over 35% of their produce value between their farmgate and the terminal market. Why? Because middlemen arbitrarily downgrade produce quality, transportation is unorganized and expensive, and payments take 30 to 45 days.

We built **CROP2GO** — an AI-integrated ecosystem that unites the **Farmer, the FPO, the Institutional Buyer, and the Commercial Transporter** onto a single transparent railway. With computer vision produce grading, village-level shared freight aggregation, and live GPS telemetry, we elevate farmer earnings by 35% and eliminate post-harvest wastage."

The 3 Unfair Advantages

- **Automated AI Computer Vision Grading:** Replaces manual eyeballing with unbiased, multi-class quality distribution (Grades A, B, C, and Rejected).
- **Shared Demand Logistics Aggregator:** Groups nearby village farmers booking small tonnage on the same date to fill a single 5-ton or 10-ton reefer truck, cutting freight costs by 42%.
- **Zero-Waste Value Pathway:** Grade C produce is automatically routed to agro-processing units (puree, ketchup, dehydration), turning waste into cash.

The 4-Actor Ecosystem Credential Matrix

Role	Demo Phone	Password	Showcase Focus
FPO Admin	9999900001	password123	AI Grading & Master Lots
Lead Farmer	9999900010	password123	My Crop & Payout Passbook
Wholesale Buyer	9999900020	password123	Master Order & GPS Tracking
Transporter	9999900040	password123	Device Telemetry Broadcast

Section 2: The Interconnected SIH Presentation Storyline

Every data record currently seeded in CROP2GO belongs to a single cohesive supply chain journey from the fertile soil of **Baramati, Pune** to the wholesale hypermarket at **Vashi APMC, Navi Mumbai**.

The Single Batch Traceability Chain: Batch Code #TOM-0908

STAGE	ENTITY / CODE	KEY ATTRIBUTES & VALUES	INTERCONNECTED LINK
1. Farm Production	CRP-2026-TOM-01	Farmer: Ramesh Patel (Village Rui) Variety: Abhinav Hybrid, Area: 3.5 Acres Harvest: 6,000 kg Tomatoes (Day 75)	Produced by Farmer 9999900010 on Ganga Plot Khasra 42. Tracked with ₹55,500 expenses.
2. Shared Logistics	SLOT-TOM-9081	Produce: Tomato, Qty: 6,000 kg Pickup: Village Rui, Baramati Assigned: Eicher Pro Reefer (MH 12 AB 9021)	Aggregated with Village Katewadi farmers (SLOT-TOM-9082 : 4,000 kg & 9083 : 2,500 kg = 12,500 kg cluster).
3. FPO Weighing	TOM-0908-01	Gross Net Weighed: 5,850 kg Center: Baramati Central Hub	Digital weighing scale record created by FPO Worker Mohan Sharma (9999900002).
4. AI Quality Grading	GRADE A: 60% GRADE B: 30% GRADE C: 8% REJ: 2%	AI Confidence: 96% Grade A: 3,510 kg (Export/Retail) Grade B: 1,755 kg (Mandi Bulk) Grade C: 468 kg (Processing Puree) Rejected: 117 kg	Grading record linked to Lot TOM-0908-01 . Combined with Suresh & Anand's lots to form Master Lot.
5. Master Lot Aggregation	ML-TOM-PUN-01	Crop: Tomato, Grade: Grade A Premium Total Aggregated: 6,900 kg Contributing Farmers: 3	Aggregates Grade A produce from: Ramesh (3,510 kg), Suresh (2,156 kg), Anand (1,240 kg).
6. Processing Pathway	ML-TOM-PROC-03	Grade: Grade C Puree Quality Quantity: 980 kg Buyer: Sahyadri Food Processing Ltd	Sold to processor at ₹14/kg (ORD-SIH-9082) recovering ₹13,720 value. Zero food waste!
7. Institutional Contract	ORD-SIH-9081	Buyer: Reliance Fresh Agri Wholesale	Buyer Vikramaditya Singhania (9999900020) purchases from Master Lot ML-TOM-PUN-01 . Paid in

STAGE	ENTITY / CODE	KEY ATTRIBUTES & VALUES	INTERCONNECTED LINK
		Quantity: 6,000 kg Grade A Price: ₹38.00 / kg Total Contract Value: ₹2,28,000	full escrow.
8. Live GPS Dispatch	DISP-SIH-7701	Vehicle: MH 12 AB 9021 (Reefer Truck) Driver: Ramesh Patil (9999900040) Temp: 11.5°C , Speed: 54 km/h Position: Khandala Ghat, Mumbai-Pune Exp	Transporter streams real device browser telemetry via watchPosition . Real-time Leaflet map tracking.
9. Transparent Payouts	PAY-SIH-001	Farmer: Ramesh Patel Share: 50.8% of Master Lot Net Direct Bank Payout: ₹1,10,032	FPO retains 5% service margin (₹11,400). Payout automatically calculated & deposited in farmer digital passbook!

Section 3: Live Demonstration Script (Step-by-Step)

Follow this exact sequence on your projector or screen during the 5-minute presentation:

ACT 1: The Farmer's Portal — Intelligent Production & Shared Logistics

Login: 9999900010 | Ramesh Patel

1. Open <http://localhost:5173/login>. Click the **Farmer (9999900010)** demo button and sign in.
2. Click **My Crops** on the sidebar: Show the jury **Ganga Plot (Khasra 42)** with the active *Abhinav Hybrid Tomato* crop cycle at Day 75 (Peak Harvest, 82% progress, Excellent health).
3. Click **Expenses**: Show how Ramesh tracked ₹55,500 of input costs (Drip, Seeds, Labour).
4. Click **Transport**: Show how Ramesh and neighboring farmers from Village Rui grouped together under **SLOT-TOM-9081** to book shared village freight.

Script to say: "Jury members, look at Ramesh's screen. As an individual smallholder farmer with 6 tons of tomatoes, hiring an entire truck to Mumbai would cost him ₹9,000. Through our village clustering module, his shipment is combined with Suresh and Anand's harvest, reducing his logistics expense to just ₹4,200 — an instant 42% cost saving!"

ACT 2: The FPO Operational Engine — Digital Weighing & AI Computer Vision Grading

Login: 9999900001 | Rajesh Kumar (FPO Admin)

1. Logout and click the **FPO Admin (9999900001)** demo button to sign in.
2. Navigate to **Collection & Weighing**: Show Lot **TOM-0908-01** brought by Ramesh Patel, digitally weighed at **5,850 kg**.
3. Navigate to **AI Grading**: Select Lot **TOM-0908-01**. Click *AI Assist Quality Grading*.
4. Highlight the AI breakdown: **60% Grade A (3,510 kg)**, **30% Grade B**, **8% Grade C**, and **2% Rejected** with a 96% AI confidence rating.
5. Navigate to **Aggregation**: Show Master Lot **ML-TOM-PUN-01** aggregating **6,900 kg of Grade A tomatoes** from 3 member farmers.
6. Navigate to **Processing**: Show the 980 kg of Grade C tomatoes being routed to *Sahyadri Food Processing Ltd* for ketchup/puree.

Script to say: "In traditional mandis, commission agents glance at a crate and classify the entire truckload as Grade B, pocketing the difference. With our AI Grading system, the produce is objectively categorized. 60% qualifies for premium export and supermarket pricing. Crucially, the Grade C tomatoes aren't dumped into landfills — our Decision Intelligence routes them directly to food processors at ₹14/kg, ensuring zero wastage!"

ACT 3: The Transporter Portal — Live Browser GPS Telemetry on Highway

Login: 9999900040 | Ramesh Patil (Transporter)

1. Logout and click the **Transporter (9999900040)** demo button to sign in.
2. Show the Transporter Dashboard with verified vehicle plate **MH 12 AB 9021** and commercial DL number.
3. Under **Live GPS Transmitter**, trip **DISP-SIH-7701** is selected with destination *Reliance Fresh Hub, Vashi APMC Terminal*.
4. Click **Start Trip & Broadcast Device GPS** (or click **Send Single Telemetry Packet** for desktop test).
5. Show the real-time telemetry streaming: Speed **54 km/h**, Accuracy **±6 meters**, Lat **18.7562°N**, Lng **73.3718°E** near Khandala Ghat.

Script to say: "Notice something crucial: The driver didn't have to purchase a ₹15,000 telematics OBD box! Using the modern HTML5 Geolocation API, any ordinary smartphone in the driver's pocket broadcasts sub-10-meter live coordinates, speed, and heading straight to the FPO and Buyer dispatch centers!"

ACT 4: The Institutional Buyer — Real-Time Route Tracking & Verification

Login: 9999900020 | Reliance Fresh Procurement

1. Logout and click the **Buyer (9999900020)** demo button to sign in.
2. Under **My Orders**, show Order **ORD-SIH-9081**: 6,000 kg Grade A Tomatoes purchased at ₹38/kg (Total: ₹2,28,000).
3. Click **Track Shipments**: Behold the interactive **Leaflet OpenStreetMap**!
4. Point out the live truck icon traveling along the Mumbai-Pune Expressway, with origin at Pune FPO, current checkpoint at Khandala, destination at Vashi APMC, active cold-chain temperature (11.5°C), and an estimated arrival in 1 hr 25 mins.

Script to say: "Institutional buyers like Reliance Fresh or BigBasket pay a premium because they receive guaranteed Grade A produce with 100% digital traceability. They can see the temperature history inside the refrigerated container and know exactly when the truck will dock at their receiving bay."

ACT 5: Closing the Loop — Direct, Audited Farmer Payout Distribution

Login: 9999900010 | Ramesh Patel

1. Log back in as **Farmer Ramesh Patel (9999900010)**.
2. Click **Payments** on the sidebar.
3. Show Payment Voucher **PAY-SIH-001**: Total Earned **₹1,10,032**.
4. Click to expand the audit breakdown: 3,510 kg Grade A Tomatoes contributed to Master Lot **ML-TOM-PUN-01**, sold at ₹38.00/kg (Gross ₹1,33,380) minus 5% FPO facilitation fee, deposited directly into his bank account via NEFT.

Script to say: "And this is the climax of CROP2GO. The money didn't get stuck in a trader's ledger for 45 days. The smart distribution algorithm calculated Ramesh's exact 50.8% share of the batch and credited ₹1,10,032 into his account within 24 hours. Full transparency, zero leakages, and zero middlemen."

Section 4: Anticipated Jury Questions & Winning Answers

Q1: "How does your AI grading work if lighting conditions change in a rural mandi or farm?"

Winning Answer: "Excellent question, Sir. Our AI computer vision model utilizes adaptive histogram equalization and HSV color-space normalization before running convolutional inference. This ensures that ambient shadow variations or yellow incandescent bulbs do not distort surface defect detection or ripeness classification. Additionally, the FPO worker retains manual override capabilities with an audit trail, keeping human expertise in the loop."

Q2: "What if the driver travels through rural areas with no mobile internet connectivity?"

Winning Answer: "Our driver web app uses client-side IndexedDB local caching with service workers. When cellular connectivity drops in a ghat or rural blind spot, the browser continues logging hardware GPS fixes with timestamps locally. The moment a 3G/4G signal is regained, the packet queue auto-synchronizes to the server, preserving the complete continuous route history without data loss."

Q3: "How does CROP2GO differ from Government e-NAM?"

Winning Answer: "e-NAM is primarily an auction bidding gateway for physical APMC mandis. It does not handle village-level logistics aggregation, does not offer computer vision quality grading, and does not provide real-time highway telematics. CROP2GO complements e-NAM by serving as the *pre-mandi aggregation and fulfillment operating system* for FPOs, enabling them to trade on e-NAM or sell directly to corporate contract buyers with verified digital lot passports."

Q4: "What is the economic sustainability and business model for the FPO?"

Winning Answer: "The FPO earns through a nominal 3% to 5% facilitation fee on completed buyer contracts (as demonstrated in Order ORD-SIH-9081 where the FPO earned ₹11,400). Because farmers achieve a 30-40% higher realization through AI grading and master lot contracts compared to unorganized distress sales, they willingly celebrate the 5% FPO margin. The FPO also earns recurring revenue from shared drone spraying and machinery hiring."

Section 5: SIH Presentation Quick-Reference Cheat Sheet

ROLE / PERSONA	LOGIN PHONE	PASSWORD	ASSOCIATED BATCH / RECORD	SCREEN / URL PATH
FPO Admin (Rajesh Kumar)	9999900001	password123	FPO ID 1: Sahyadri Kisaan Samriddhi	/fpo (Collection, Grading, Dispatch)
Lead Farmer (Ramesh Patel)	9999900010	password123	Lot TOM-0908-01, Payout PAY-SIH-001	/farmer (My Crops, Payments, Transport)
Wholesale Buyer (Reliance Fresh)	9999900020	password123	Order ORD-SIH-9081 (₹2,28,000)	/buyer (My Orders, Track Shipments)
Transporter (Ramesh Patil)	9999900040	password123	Truck MH 12 AB 9021, Dispatch 7701	/transport (GPS Transmitter, Dispatches)

